

MTO-Net Literature Review and Novelty Audit

Executive assessment

Bottom-line verdict. MTO-Net looks **scientifically interesting and potentially publishable at a high level**, but only if it is positioned as a **new equivariant atom-to-molecule assembly framework for learning explicit, delocalized latent tensor modes** rather than as a replacement for quantum chemistry, a discovery of “neural molecular orbitals,” or a general solution to “computable chemical intuition.” The strongest novelty is **not** the use of equivariance, tensor features, spectroscopy targets, or orbital language individually; all of those already exist. The plausible novelty is the **combination** of these ingredients into an explicit **multi-mode, molecule-level, equivariant latent representation** that is trained **without Hamiltonian, density-matrix, orbital-coefficient, or wavefunction supervision**, and then shown to be **stable, transferable, and chemically reusable**. Existing work comes close on individual components, but I did **not** find a clear prior molecular method that already does that full package. ¹

Conceptual originality. If the paper claims that MTO-Net introduces a **learned molecular response mode basis** assembled from atom-centered irreducible tensor features, then the conceptual originality is **moderate-to-high**. If, instead, it claims to inaugurate an entirely new “chemical language,” or contrasts itself too sharply against all graph neural networks via a blanket “valence-bond-like versus molecular-orbital-like” dichotomy, the originality becomes less defensible because the literature already contains powerful equivariant models, orbital-informed models, graph pooling/global-token methods, tensorial-property predictors, and even recent equivariant orbital-learning work. ²

Likely closest prior art. If forced to name a single closest conceptual neighbor, I would choose **CEONET** because it is the most direct combination of **equivariance, orbital-centered scientific framing, and molecular-orbital representation learning**. But CEONET is also scientifically distinct, because it learns on or reconstructs **actual orbital objects** and derived quantum-chemical quantities, whereas MTO-Net is proposed to learn **latent response modes** without electronic-structure supervision. In architectural spirit for spectroscopy and tensorial-response tasks, **DetaNet** and especially **EnviroDetaNet** are also very close. In graph-architecture terms, the most relevant non-chemistry novelty threat is **equivariant hierarchical pooling** such as **EGHN**, because skeptical reviewers may view MTO-Net as a chemically branded version of equivariant pooling unless the mode bank is shown to be more than that. ³

Strongest narrative. The strongest narrative is **Narrative 4**: that MTOs may form a **stable, reusable molecular response representation**. This is scientifically testable and potentially impactful if supported by multi-seed alignment, cross-task transfer, frozen-probe utility, fragment enrichment, and intervention studies. The weakest narrative is **Narrative 1** in its broadest form, because “making chemical intuition computable” is rhetorically attractive but too expansive unless narrowed to specific computable priors and measurable representational outcomes. ⁴

Largest novelty risk. The biggest risk is that reviewers will say: this is an equivariant readout / pooling / latent-slot mechanism with orbital rhetoric. That objection will be hard to answer unless MTO-Net demonstrates properties that ordinary pooling, global tokens, attention heads, or hierarchical clustering do not usually provide: a **fixed, explicit mode bank, equivariant transformation laws at the mode level, cross-task reuse, and seed-stable subspaces with chemically interpretable interventions**. ⁵

Largest scientific opportunity. The upside is substantial. If MTO-Net can show that molecule-level tensor modes emerge as reproducible, delocalized latent structures that generalize across spectroscopic and response-property tasks, then it would offer a useful middle ground between atomwise GNN embeddings and fully supervised electronic-structure learning. That would likely matter most to **chemical machine learning / AI for science**, and secondarily to **spectroscopy** and **theoretical chemistry** if the interpretation evidence is unusually strong. ⁶

Direct answers to the requested questions

Question	Answer	Confidence
What is the single closest existing method?	CEONET is the closest conceptual prior art; EnviroDetaNet is the closest task/application prior art; EGHN is the closest architectural prior art. If forced to choose one, I would name CEONET . ⁷	Medium
Has any previous model constructed multiple explicit molecule-level equivariant modes from atom-centered tensor fields?	I found no clear molecular precedent that explicitly constructs a fixed bank of multiple molecule-level equivariant latent modes from atom-centered tensor fields with a role close to MTOs . Existing work provides global tokens, pooling clusters, hierarchical coarsening, or real-orbital prediction instead. ⁸	Medium
Is MTO-Net a new representation class or mainly a new pooling/readout mechanism?	As formulated on paper alone, mostly a new pooling/readout family. It becomes a new representation class only if the learned modes are shown to be explicit, stable, reusable, and distinct from generic global embeddings. ⁹	High
What distinguishes an MTO from an attention head, latent slot, graph token, or pooled embedding?	Only a combination of: explicit mode bank, O(3)-equivariant tensor transformation laws, cross-order assembly from irreps, delocalized support, and reusability across tasks . Without those, the distinction is weak. ¹⁰	High
Is the graph/VB comparison defensible?	Loosely, yes; strongly, no. Molecular graphs do encode localized atoms-and-bonds languages reminiscent of classical structural chemistry, but modern message passing already captures nonlocal effects through depth, attention, virtual/global tokens, and geometric interactions. ¹¹	High
How should the LCAO analogy be phrased?	As an architectural analogy : MTO-Net assembles learned molecule-level tensor modes from atom-centered tensor bases, inspired by LCAO, without implying they are molecular orbitals, eigenvectors, or wavefunctions. ¹²	High
Is “response mode” the best term?	It is usable, but safest as “learned latent molecular response mode” or “neural molecular response mode.” Unqualified “response mode” risks collision with established physics/chemistry terminology. ¹³	Medium

Question	Answer	Confidence
What evidence would justify “stable, reusable, chemically meaningful” ?	Aligned multi-seed subspace stability, frozen-probe transfer across response tasks, mode-intervention effects, fragment enrichment against matched nulls, and performance beyond generic pooling baselines. ¹⁴	High
Does seed-level subspace stability show uniqueness or only functional equivalence?	Usually not uniqueness . It mainly shows reproducibility of a subspace or coding geometry , often after alignment. That is weaker than identifiability of individual modes. ¹⁵	High
Most serious conceptual objections?	“It is just equivariant pooling” ; “orbital language is misleading” ; “enrichment is not mechanism” ; “symmetry does not imply physical meaning” ; “O(3) may be wrong for chirality-sensitive tasks.” ¹⁶	High
What part is genuinely novel?	The potentially novel part is the explicit molecule-level equivariant mode bank plus evidence that it is stable and reusable . The induction/deduction framing, symmetry arguments, and orbital inspiration are mostly background or positioning. ¹⁷	High
Minimum Nature-level evidence?	More than accuracy: at least multi-task generality, strong ablations against pooling/global-token controls, seed stability, frozen transfer, intervention-based interpretability , and ideally some experimentally anchored chemical insight . ¹⁸	Medium
Stronger evidence for “new scientific language” ?	Cross-dataset and cross-lab reuse, prospective hypothesis generation, and mode manipulations that predict new response changes later confirmed computationally or experimentally.	Medium
Which community cares most?	Primarily chemical machine learning / AI for science ; secondarily spectroscopy ; only then theoretical chemistry , unless the work shows unusually careful non-overclaiming physical interpretation. ¹⁹	High
Best journal targets?	Highest-evidence tier: Nature Chemistry, Nature Machine Intelligence, Nature Computational Science. Strong but more specialized tier: Nature Communications, JACS, JCTC, Chemical Science, Digital Discovery.	Medium

Historical and conceptual background

Chemical intuition, reduction, and computability

Chemistry has always operated with layered explanatory languages. At one level are quantum mechanics and electronic structure; at another are practical, reduced concepts such as functional groups, aromaticity, resonance, donor–acceptor interactions, polarization, steric effects, motifs, and fragments. Modern reviews on molecular representation learning and geometric deep learning emphasize exactly this plurality of representations, from handcrafted descriptors to learned 3D geometric embeddings. The literature on functional groups and aromaticity likewise shows that chemists rely on reduced concepts because they are compact, transferable, and experimentally useful—even when those concepts are not uniquely derivable from first principles in a simple one-to-one way. ²⁰

That makes the phrase “**making chemical intuition computable**” defensible only in a **narrow methodological sense**: one can make selected reduced chemical regularities computable by encoding fragment structure, local environments, orbital interactions, or concept bottlenecks in a machine-learning pipeline. Group-contribution methods, descriptor-based QSAR, expert systems for synthesis planning, and interpretable fragment-aware GNN models are all precedents for this move. But the phrase becomes too broad if it implies that a model has captured chemistry’s full intuitive language or that philosophical motivation substitutes for empirical evidence. ²¹

So Narrative 1 should be treated as **motivation**, not proof. The safe framing is: chemists often use reduced conceptual languages before invoking full quantum calculations; MTO-Net attempts to learn a reusable symmetry-constrained latent representation that may serve as a computable analogue of some of those reduced response concepts. That is a much narrower and more defensible ambition than claiming to formalize “chemical intuition” wholesale. ²²

Valence-bond and molecular-orbital languages

The history of valence-bond and molecular-orbital theory does support careful analogy, but only careful analogy. Modern retrospectives stress that VB and MO are not simply “wrong versus right”; they are complementary quantum-mechanical formalisms emphasizing localized bonding pictures versus delocalized molecular states. Recent chemistry education and historical essays explicitly argue that the common textbook framing of MO as superior and VB as merely approximate is misleading. ²³

That matters for MTO-Net because ordinary molecular graphs and atom–bond GNNs can indeed be described as **loosely VB-like** in the sense that they start from localized atoms, bonds, and neighborhoods. But this is an **interpretive analogy**, not a historical lineage: message passing, continuous-filter convolutions, directional interactions, and equivariant attention are mathematical constructions in machine learning, not descendants of Heitler–London or Pauling theory. Reviewers will likely accept “loosely analogous to localized structural chemistry” and resist “inherits valence-bond theory” unless the language is softened. ²⁴

The MO/LCAO analogy is more useful for MTO-Net if it is formulated as follows: **atom-centered equivariant basis-like features are assembled into explicit molecule-level, delocalized latent modes**. That is close enough to LCAO to be scientifically illuminating, provided the paper repeatedly makes clear that these are **not** electronic eigenstates, Kohn–Sham orbitals, or wavefunctions unless separately demonstrated. CEONET, SchNOrb, PhiSNet, OrbNet-Equi, and Orbital Mixer show that the literature already uses orbital language in machine learning, but usually either with real electronic-structure targets or with quantum-derived orbital features. MTO-Net’s distinction would be the absence

of that supervision and the shift from **electronic structure prediction** to **latent response representation learning**. ²⁵

Symmetry as axiom, not mechanism

Symmetry is the most solid part of the conceptual foundation. Tensor Field Networks, symmetry-adapted tensorial-property learning, PaiNN, NequIP, MACE, Equiformer, FIREANN, and the recent JACS framework for tensorial properties all build on the idea that molecular observables must transform correctly under rotations, translations, permutations, and sometimes inversion/parity. This has direct relevance to dipoles, multipoles, polarizabilities, Raman tensors, NMR shielding tensors, and spectroscopy. ²⁶

But symmetry constraints alone do **not** justify calling a latent representation physically meaningful. They guarantee transformation behavior, not mechanistic identity with quantum objects. That distinction is crucial. SchNOrb, PhiSNet, QHNet, HamGNN, DeepH, CEONET, FermiNet, and PauliNet all connect more directly to quantum-mechanical states, operators, or variational principles than the proposed MTO-Net does. MTO-Net can borrow **symmetry** and **compositionality** from physics, but it does not solve the Schrödinger equation and should not imply otherwise. ²⁷

O(3), parity, and chirality

The choice between SO(3), O(3), E(3), and parity-sensitive formulations must be handled carefully. O(3)-equivariant models naturally encode reflection behavior, which is often desirable for ordinary tensorial properties. But chirality-sensitive tasks are a limitation zone: recent work argues that standard E(3)-equivariant/invariant models cannot properly learn chirality from ordinary 3D atom-point representations alone, and specialized alternatives such as ChIRo and geometry-complete/chirality-aware architectures were introduced precisely because stereochemistry can be lost under common symmetry assumptions. This means MTO-Net should not claim general physical completeness under O(3) without clarifying which target properties are parity-even, parity-odd, or chirality-sensitive. ²⁸

Literature map for the four narratives

From chemical induction to computable chemical intuition

The literature supports the motivation but not the grandiose formulation. Descriptor-based QSAR, group-contribution models, fragment-based reasoning, computer-aided synthesis planning, and concept-bottleneck approaches all show long-standing attempts to formalize reduced chemical knowledge. Recent interpretable GNN work makes that tradition more expressive by learning fragment-level or concept-level predictors rather than relying only on hand-designed features. ²¹

The limit is equally clear. The interpretable-ML literature repeatedly shows that attribution, concept bottlenecks, or chemically plausible fragments do not automatically yield faithfulness or mechanism. Molecular explanation papers and benchmark studies warn against reading substructure importance or latent concept activation as proof of causal chemical reasoning. Therefore, Narrative 1 can motivate why one might want MTOs, but cannot serve as evidence that MTOs are real, stable, or physically meaningful. ²⁹

Skeptical synthesis: Narratively valuable, scientifically secondary. Keep it as a framing device and quickly move to the concrete gap: existing molecular ML lacks an explicit, reusable, equivariant molecule-level response representation learned without electronic-structure supervision. ³⁰

Valence-bond-like graph language versus molecular-orbital-inspired assembly

This comparison is the most rhetorically tempting and the most dangerous. Ordinary molecular GNNs do begin from localized atoms and edges, which makes a loose analogy to localized chemical languages reasonable. But modern models already propagate nonlocal information through directional message passing, spherical harmonics, dipole/vector channels, global attention, and long-range geometric interactions. DimeNet, GemNet, PaiNN, Equiformer, ViSNet, and TorchMD-Net all go well beyond static local bond labels. ³¹

So MTO-Net cannot defend novelty by implying that all previous graph models are purely local or intrinsically incapable of delocalization. They are not. The stronger and fairer distinction is this: previous models usually **encode** long-range effects implicitly inside atomwise features, attention maps, or a single pooled graph embedding, whereas MTO-Net is supposed to **materialize multiple explicit molecule-level tensor modes** that can be individually analyzed, aligned across seeds, transferred across tasks, and related to chemistry. If that explicit mode materialization is real, it is a meaningful distinction from ordinary pooling. If not, the MO analogy collapses into branding. ³²

The literature also narrows the novelty window from the other side. OrbNet-Equi, SchNOrb, PhiSNet, Orbital Mixer, QHNet, HamGNN, and CEONET already use orbital or Hamiltonian language, often with symmetry constraints. Thus, MTO-Net must emphasize the point of separation: **inspiration from orbital assembly without claiming to predict, reconstruct, or approximate physical orbitals**. ³³

Skeptical synthesis: The VB/MO contrast is tenable only as a **qualified architectural analogy**. It should never be the sole novelty claim.

Symmetry as an axiom for an independent neural description of chemistry

Narrative 3 is well grounded in established literature. Symmetry-aware models have repeatedly shown benefits for atomistic learning, particularly for vector and tensor observables. Symmetry-adapted tensorial-property learning, PaiNN, FIREANN, DetaNet, EnviroDetaNet, and the 2025 JACS tensorial-property framework all make the case that irreducible tensor channels are natural latent carriers for response properties and spectra. ³⁴

Where this narrative becomes vulnerable is when it drifts from “correct covariance laws” to “physically meaningful internal representation.” Recent probing work on atomistic foundation models already shows property information routed into certain irreps or channels, which means that mere irrep-specific routing is no longer by itself a strong novelty claim. Moreover, symmetry-respecting models can still learn arbitrary task-driven latent bases, and a representation may be symmetry-correct while remaining scientifically opaque. ³⁵

Skeptical synthesis: Symmetry is an essential axiom for MTO-Net, but it is **table stakes**, not sufficient novelty. The paper should present “symmetry-constrained neural chemistry” as a **proposed methodological program inside geometric deep learning**, not as an already established field or as proof of physical validity. ³⁶

Stable, reusable, and chemically meaningful molecular response modes

This is the make-or-break narrative. The representation-similarity literature has matured enough that seed stability can no longer be asserted using only similar predictions or visual overlap. SVCCA, PWCCA, CKA, orthogonal Procrustes alignment, grounded similarity testing, and principal-angle analysis provide

a serious toolkit for comparing latent representations across random initializations. That literature also warns that representations can differ by basis change, permutation, sign, or isotypic rotation while remaining functionally equivalent. ³⁷

The chemical-interpretability literature raises a second bar. Substructure masking, concept whitening, group-contribution hybrids, and counterfactual perspectives show how chemically plausible explanations can be produced; quantitative benchmarks show how often such explanations are unstable, unfaithful, or dataset-contingent. So an enrichment map around “chemically meaningful fragments” is not enough. MTO-Net would need intervention studies showing that editing a mode or projecting it out changes predicted response behavior in chemically coherent ways, beyond what generic embeddings or attention heads can already do. ³⁸

A further novelty pressure comes from very recent atomistic foundation-model probing: irreps-specific routing and linear/frozen-probe utility are already being analyzed as reusable latent structure in 2026 preprints. That means MTO-Net must go beyond merely showing that some channels linearly encode dipoles or spectra. It must show that **explicit molecule-level modes** are better probes, more stable, or more interpretable than the standard hidden states of existing atomistic models. ³⁹

Skeptical synthesis: This is the strongest and most novel narrative, but also the one that requires the most disciplined evidence.

Taxonomy of related methods and closest prior art

Field taxonomy and where MTO-Net would sit

Category	What the literature already contains	Where MTO-Net would sit
Equivariant molecular neural networks	Atom-centered scalar/vector/tensor message passing and equivariant attention for energies, forces, and molecular properties. ⁴⁰	Uses the same atom-centered equivariant backbone as a starting point, but claims a new molecule-level mode assembly stage.
Electronic-structure and wavefunction learning	Direct prediction of Hamiltonians, overlap matrices, orbitals, densities, and wavefunctions. ⁴¹	Distinct if it does not require such labels and does not predict physical orbitals or Hamiltonians.
Orbital-informed machine learning	Uses atomic-orbital features or semiempirical QM inputs to improve chemistry prediction. ⁴²	Adjacent in inspiration, but different if MTO-Net learns latent modes from geometry and atom types alone .
Neural orbitals in variational quantum mechanics	Neural parametrizations of many-electron wavefunctions and orbitals under first-principles variational training. ⁴³	Scientifically separate problem class.
Global and hierarchical molecular representations	DiffPool, graph multiset pooling, global tokens, hierarchical equivariant pooling, multimodal/foundation representations. ⁴⁴	This is the largest novelty threat if MTOs are not shown to be more than explicit equivariant pooling/slots.

Category	What the literature already contains	Where MTO-Net would sit
Tensorial properties and spectroscopy	Dipoles, polarizabilities, field response, spectra from equivariant intermediate representations. ⁴⁵	MTO-Net naturally belongs here first.
Interpretable chemistry and concept discovery	Substructure masking, concept whitening, fragment-level GNNs, explanation benchmarks, counterfactual perspectives. ⁴⁶	MTO-Net belongs here only if mode meaning is demonstrated by interventions, not just pictures.
Foundation models and reusable molecular representations	Large-scale pretraining, multimodal molecular representations, equivariant pretrained transformers. ⁴⁷	MTO-Net could become a reusable head or representation family within this trend.

Closest-prior-art comparison table

Legend. QC in/labels : what extra quantum information is required at train or test time. Scope : A atom-centered, G single graph/global embedding, H hierarchical/global coarsening. Latent : S scalar, T tensor/irreps. Modes : explicit multiple molecule-level modes? DeLoc : explicitly delocalized mode objects? Orb-like : intentionally orbital-like? ES target : predicts real orbitals / Hamiltonian / wavefunction? Stab/Transfer/Interp : explicit analysis in paper. Overlap : D0 direct overlap; C0 component overlap; CA conceptual adjacency. Threat is threat to MTO-Net novelty, not paper quality.

Paper	Year and venue	QC in/labels	Scope	Latent	Modes	Deloc	Orb-like
Tensor Field Networks	2018, arXiv	None	A	T	N	N	N
Cormorant	2019, NeurIPS	None	A/G	T	N	Implicit	N
SE(3)-Transformer	2020, NeurIPS	None	A/G	T	N	Implicit	N
DimeNet	2020, ICLR	None	A/G	mixed directional	N	Implicit	N
GemNet	2021, NeurIPS	None	A/G	mixed directional	N	Implicit	N

Paper	Year and venue	QC in/labels	Scope	Latent	Modes	Deloc	Orb-like
PaiNN	2021, ICML	None	A/G	T	N	Implicit	N
NequIP	2022, Nat. Commun.	None	A/G	T	N	Implicit	N
MACE	2022, NeurIPS	None	A/G	T	N	Implicit	N
Equiformer	2023, ICLR	None	A/G	T	N	Implicit	N
ViSNet	2024, Nat. Commun.	None	A/G	T	N	Implicit	N
DetaNet	2023, Nat. Comput. Sci.	None	A/G	T	N	Implicit	N
EnviroDetaNet	2025, npj Comput. Mater.	None	A/G/H	T	N	Implicit	N
FIREANN	2023, Nat. Commun.	None	A/G	T	N	Implicit	N
DiffPool	2018, NeurIPS	None	H	S	Partial	cluster-level	N

Paper	Year and venue	QC in/labels	Scope	Latent	Modes	Deloc	Orb-like
Graph Multiset Pooling	2021, ICLR	None	G	S	Partial	latent global slots	N
EGHN	2022, NeurIPS	None	H	equivariant coordinates/features	Partial	cluster-level	N
OrbNet-Equi	2022, PNAS	semiempirical AO features	A/G	T	N	implicit electronic	weak
SchNOrb	2019, Nat. Commun.	orbital matrices labels	A/G	mixed	N	physical electronic structure	strong
PhiSNet	2021, NeurIPS	wavefunction/density labels	A/G	T	N	physical electronic structure	weak
QHNet	2023, ICML	Hamiltonian labels	A/G	T	N	physical electronic structure	weak
HamGNN	2023, npj Comput. Mater.	Hamiltonian labels	A/G	T	N	physical electronic structure	weak
Orbital Mixer	2022, JCTC	atomic orbital features and labels	G	basis-dependent	N	physical orbital space	strong

Paper	Year and venue	QC in/labels	Scope	Latent	Modes	Deloc	Orb-like
CEONET	2025, PNAS	real orbital information / orbital-centric supervision	G	T	not latent bank; real-orbital objects	Y	Y
FermiNet	2020, Phys. Rev. Research	none, variational first-principles	global electronic wavefunction	antisymmetric neural orbitals	Y, but physical ansatz	Y	Y
PauliNet	2020, Nat. Chem.	none, variational first-principles	global electronic wavefunction	antisymmetric neural orbitals	Y, but physical ansatz	Y	Y
Uni-Mol	2023, ICLR	none	G	mostly scalar geometric transformer	N	implicit	N
EPT	2026, Nat. Commun.	none	G	T	one or few global transformer tokens	N/Partial	N

Overall novelty audit from the table. I do **not** see a convincing **direct overlap** paper. I do see several **component overlaps** that are strong enough to force careful positioning: **CEONET** for orbital-style equivariant molecular objects; **DetaNet/EnviroDetaNet/FIREANN/JACS 2025 tensor framework** for tensorial-response and spectroscopy; **EGHN/DiffPool/GMT/EPT** for the possibility that reviewers will reclassify MTO-Net as enhanced pooling or global-token assembly. ⁷⁵

Novelty and claim–evidence matrix

Claim audit

Proposed claim	Relevant supporting literature	Relevant limiting literature	Status	Evidence MTO-Net itself must provide	Strongest defensible wording	Wording to avoid
MTO-Net makes inductive chemical intuition computable	group-contribution, concept bottlenecks, fragment-aware GNNs, interpretable chemistry ML ⁷⁶	explanation-faithfulness critiques; CBM limitations ⁷⁷	Plausible but broad	define which “intuition” is targeted; show computable response concepts, not just good predictions	“MTO-Net operationalizes selected reduced chemical regularities as explicit latent response modes.”	“MTO-Net formalizes chemical intuition”
Conventional molecular GNNs inherit a valence-bond-like atom-bond language	graph-based molecular representations start from atoms/bonds; local structural encodings are standard ⁷⁸	modern GNNs already encode nonlocal geometric information and global context ⁷⁹	Plausible as analogy	none, but wording must be qualified	“Many molecular GNNs begin from a localized atom-bond representation loosely analogous to classical structural chemistry.”	“Molecular GNNs are valence-bond models”
MTO-Net inherits the MO idea of assembling molecular objects from atomic bases	LCAO/MO analogy; orbital-informed ML; CEONET/OrbNet-Equi show scientific relevance of orbital assembly ideas ⁸⁰	orbital language is already used elsewhere and can imply physical orbitals if unqualified ⁸¹	Plausible	architecture diagram and equations showing explicit atom→molecule mode assembly	“MTO-Net is inspired by the LCAO principle of assembling molecule-level objects from atom-centered bases.”	“MTO-Net neural molecular orbitals”
MTOs are learned delocalized molecular response modes	spectra/response models motivate response-oriented latent structure ⁸²	current literature already has implicit long-range representations without explicit modes ⁸³	Plausible, currently unproven	atom-support spread, participation ratio, fragment coverage, long-range sensitivity, knockouts	“MTOs are explicit learned molecule-level latent tensor modes with delocalized support.”	“MTOs are response eigenmodes”

Proposed claim	Relevant supporting literature	Relevant limiting literature	Status	Evidence MTO-Net itself must provide	Strongest defensible wording	Wording avoid
MTOs are not quantum-chemical orbitals or wavefunctions	distinction from SchNOrb/ PhiSNet/ QHNet/ HamGNN/ FermiNet/ PauliNet ⁸⁴	rhetorical orbital imagery may still mislead readers	Established if clearly defined	explicit disclaimer; no orbital/wf equivalence claims	“MTOs are latent neural response objects, not electronic eigenstates.”	“orbital surrogate,” “wave-like,” “directly validate”
MTO-Net captures local information and diffuse molecule-wide response	modern equivariant models carry local geometry and global interaction information ⁸⁵	previous models already do this implicitly ⁸⁶	Plausible	compare to atomwise plus ordinary pooling; show improved long-range tasks and explicit mode support	“MTO-Net jointly represents local equivariant atomic structure and assembled molecule-level response patterns.”	“Only capture response”
MTOs remain stable across random initializations	representation-similarity literature provides the right tests ⁸⁷	similarity does not imply mode identity or uniqueness ⁸⁸	Unproven	multi-seed runs; alignment within irreps; principal angles/CKA/ PWCCA/ Procrustes	“MTO subspaces are reproducible across seeds up to expected alignment freedoms.”	“MTOs unique, identifiable”
MTOs transfer across response tasks	foundation/ pretraining and frozen-probe literature says transfer can be meaningful ⁸⁹	transfer in hidden states is common; need show it is stronger or more structured than baselines	Unproven	pretrained MTOs with frozen probes and task-to-task transfer matrix	“MTOs provide a transferable molecule-level response representation across related tasks.”	“universal representation”
MTOs enrich around chemically meaningful fragments	substructure masking and concept methods give precedent for fragment-level interpretation ⁹⁰	enrichment alone is not faithfulness or causality ⁹¹	Plausible but weak alone	enrichment against matched nulls, counterfactual fragment edits, intervention on modes	“Certain MTOs concentrate on fragments enriched for chemically relevant motifs.”	“MTOs the true chemical mechanism”

Proposed claim	Relevant supporting literature	Relevant limiting literature	Status	Evidence MTO-Net itself must provide	Strongest defensible wording	Wordings to avoid
MTO-Net establishes a reusable molecular response representation	current reusable molecular representations exist, but not this exact mode framing ⁴⁷	reviewers may say it is task-specific latent space	Strong claim, unproven	cross-dataset reuse, frozen probes, out-of-task transfer, compactness vs baselines	“MTO-Net is a candidate reusable response representation.”	“MTO-Net establishes a new scientific language”
MTO-Net represents a route toward symmetry-constrained neural chemistry	symmetry-aware chemistry ML is a real and active direction ³⁶	the established umbrella term is still geometric deep learning / symmetry-adapted ML	Plausible as positioning	define term, do not present as established field	“MTO-Net suggests one route toward a more symmetry-constrained form of neural chemistry.”	“We found a new field”
MTO-Net is conceptually different from property-specific equivariant GNNs, electronic-structure-learning models, and generic pretraining	there is room between these classes ⁹²	difference vanishes if MTOs are just pooling or if no reuse is shown ⁹³	Plausible if ablated	compare against paired backbones with only readout changed; separate comparisons to electronic-structure models and pretrained models	“MTO-Net occupies a middle ground between atomwise equivariant predictors and supervised electronic-structure learning.”	“category unlike a model”

The novelty boundary in one sentence

If MTO-Net is only an equivariant way to pool atomwise tensors into a handful of global vectors/tensors, it is mainly a new readout. If it yields an explicit, reusable, stable, cross-task-delocalized mode basis with scientifically interpretable interventions, it can legitimately be argued as a new representation class. ⁹⁴

Reviewer-style critique and recommended positioning

Strongest reviewer objections and what would answer them

Reviewer perspective	Strongest likely objection	What would answer it
Equivariant ML reviewer	“This is just equivariant pooling / global tokens / latent slots with chemistry-inspired naming.” ⁹⁵	Paired ablations against mean/sum pooling, attention pooling, GMT, DiffPool, global equivariant token baselines, and EGHN-style pooling, all under matched parameter budgets; plus explicit evidence that fixed modes are reusable and stable.
Quantum chemist	“You are borrowing molecular-orbital language without predicting orbitals, Hamiltonians, or wavefunctions. Why should I believe these are anything but hidden activations?” ⁹⁶	Very explicit negative definition of MTOs; careful LCAO-inspired wording; optional post hoc comparison showing non-equivalence to physical orbitals; no physical-orbital visuals unless validated.
Chemical-interpretability reviewer	“Fragment enrichment is not mechanism; latent modes are non-identifiable; explanation methods in molecular GNNs are often unstable.” ⁹⁷	Multi-seed subspace alignment; matched-null enrichment tests; counterfactual fragment edits; causal mode deletion/projection; external chemical annotations; failure-case analysis.
General Nature editor	“Where is the conceptual advance beyond better benchmark performance?”	Demonstrate a broadly reusable response representation, not just accuracy. Show cross-task transfer, interpretability with falsifiable interventions, and ideally at least one genuinely new chemical insight or experimentally anchored prediction.

Formal definition and safe positioning

One-sentence formal definition of an MTO.

A **Molecular Tensor Orbital** is an explicitly parameterized, molecule-level, $O(3)$ -transforming latent tensor mode assembled from atom-centered equivariant tensor features to summarize a delocalized pattern of molecular response for downstream prediction.

One-paragraph formal definition of MTO-Net.

MTO-Net is an $O(3)$ -equivariant molecular learning framework in which an atom-centered equivariant backbone first produces irreducible tensor features on atoms, and a dedicated atom-to-molecule assembly module then maps those local tensor fields into a finite bank of explicit molecule-level latent tensor modes, the MTOs. The architecture is inspired by the compositional logic of LCAO—assembling molecule-level objects from atom-centered bases—but MTOs are defined operationally as learned latent response representations rather than as quantum-mechanical orbitals, Hamiltonian eigenvectors, or wavefunctions. Scientific significance would rest not merely on predictive accuracy, but on

demonstrating that these modes are stable across seeds, transferable across response tasks, chemically enriched in meaningful regions, and empirically distinguishable from generic pooling or attention mechanisms.

Central claim options

Ambition level	Recommended central claim
Safest	“MTO-Net is an equivariant atom-to-molecule assembly framework that learns explicit delocalized latent tensor modes useful for tensorial and spectroscopic molecular prediction.”
Moderately ambitious	“The explicit molecule-level tensor modes learned by MTO-Net provide a more stable and transferable response representation than standard atomwise pooling baselines.”
Strongest that could become defensible	“MTO-Net learns a reusable molecular response language: a seed-stable, symmetry-constrained mode basis that transfers across response tasks and aligns with chemically meaningful fragments.”

Terminology to use and terminology to avoid

Use	Avoid
“LCAO-inspired assembly”	“neural molecular orbitals”
“latent molecular response modes”	“predicted orbitals”
“equivariant atom-to-molecule mode assembly”	“wavefunction-free quantum chemistry”
“explicit molecule-level tensor mode bank”	“electronic structure without solving QM”
“chemically enriched latent modes”	“mechanistic explanation” unless interventions support it
“candidate reusable response representation”	“new universal chemical language”

What MTO-Net is and is not

MTO-Net **is** best understood as a **symmetry-constrained geometric deep-learning architecture** for constructing molecule-level latent tensor representations from atom-centered equivariant features. It **is not** an electronic-structure solver, not a molecular-orbital predictor, not a Hamiltonian learner, and not a proof that latent neural objects are physically real simply because they rotate correctly. ⁹⁸

A practical naming note: I did **not** find an obvious prior use of the exact phrase “**Molecular Tensor Orbital**” in chemical ML, but the acronym **MTO** is already entrenched in chemistry and materials for **methanol-to-olefins** and **muffin-tin orbitals**. If the acronym is kept, the full phrase should be spelled out aggressively and probably not used bare in titles without context. ⁹⁹

Experimental implications and introduction blueprint

Experimental tests, baselines, and ablations required

Goal	Minimum experiment	Strong version for a Nature-level paper
Performance	Compare against PaiNN, ViSNet, Equiformer, DetaNet, EnviroDetaNet, FIREANN, EPT, and a strong non-equivariant/global-token baseline on identical splits. ¹⁰⁰	Multiple response families: dipoles, polarizabilities, NMR tensors, IR/Raman/UV-Vis; both in-domain and scaffold/out-of-distribution tests.
Symmetry	Numerical rotation/reflection equivariance error tests at backbone features, MTOs, and outputs.	Also include parity/chirality-sensitive sanity tests showing when $O(3)$ is appropriate and when it fails. ¹⁰¹
Local-to-global assembly	Replace assembly block with sum/mean pooling, attention pooling, one global token, GMT, DiffPool, EGHN-like coarsening. ⁹⁵	Parameter-matched ablations plus same backbone, same losses, same datasets.
Delocalization	For each MTO compute atom-support entropy, participation ratio, radius/support spread, and fragment coverage.	Show mode support shifts meaningfully with conjugation length, polarization, donor-acceptor motifs, or chromophores.
Seed stability	10–20 seeds; align MTO sets within each irrep block using permutation matching, sign normalization, and orthogonal Procrustes; report principal angles, CKA, PWCCA/SVCCA. ¹⁰²	Stability across seeds, dataset subsamples, and mild architecture changes.
Task transfer	Pretrain on one subset of response tasks, freeze MTOs, and train linear probes on held-out response tasks.	Full task-to-task transfer matrix; compare against frozen hidden states from PaiNN/MACE/ViSNet/EPT. ¹⁰³
Frozen-probe utility	Linear probes on MTOs versus pooled atom embeddings.	Also kernel probes and low-rank nonlinear probes to distinguish linearly encoded versus genuinely reusable structure.
Chemical enrichment	Test overlap/enrichment against annotated fragments, chromophores, functional groups, conjugated subgraphs, donor/acceptor motifs.	Matched-null enrichment, scaffold-stratified significance tests, and consistency across tasks and seeds.
Intervention-based interpretability	Remove or clamp single modes; measure effect on outputs and on downstream probe performance.	Counterfactual chemistry: edit fragments predicted to couple to a mode and verify expected mode/output changes.

Goal	Minimum experiment	Strong version for a Nature-level paper
Distinction from generic pooling and attention	Compare MTO bank against multiple attention heads and multi-token global embeddings, including EPT/GMT-style controls.	Show MTOs are more stable, more transferable, or more chemically localizable than these controls.
Distinction from electronic-structure prediction	Compare to SchNOrb, PhiSNet, QHNet, HamGNN, CEONET only in a separate “extra-supervision” tier. ¹⁰⁴	Demonstrate that MTO-Net reaches useful response representations without orbital/Hamiltonian labels, while remaining non-equivalent to those physical objects.

Introduction blueprint

Paragraph 1. Chemistry advances through both empirical induction and quantum-mechanical deduction; day-to-day molecular reasoning still relies heavily on reduced concepts such as functional groups, conjugation, donor–acceptor effects, and polarization. Use this paragraph to motivate the desire for a computable, reusable intermediate representation, while keeping the philosophical scope modest. ¹⁰⁵

Paragraph 2. Molecular machine learning has become powerful, especially with geometric and equivariant models, but most representations remain atomwise and are aggregated only at readout, or else are trained to reproduce full electronic structure under costly supervision. This should define the concrete gap. ¹⁰⁶

Paragraph 3. Existing equivariant networks already handle vector and tensor information and have been successful for response properties and spectra. The unresolved question is not whether symmetry matters, but whether one can learn an **explicit molecule-level response basis** rather than leaving global behavior implicit inside hidden atom features. ¹⁰⁷

Paragraph 4. Introduce the limited VB/MO analogy carefully: graph models often start from a localized atom–bond language, whereas MTO-Net assembles molecule-level objects from atom-centered tensor bases in an LCAO-inspired—but not quantum-orbital—manner. Explicitly caution that the analogy is architectural, not ontological. ¹⁰⁸

Paragraph 5. Formally define MTO-Net and MTOs. Emphasize $O(3)$ equivariance, finite mode bank, cross-order tensor assembly, no orbital/Hamiltonian/wavefunction supervision, and intended domain of tensorial and spectroscopic response tasks. ¹⁰⁹

Paragraph 6. State the paper’s real scientific hypothesis: explicit molecule-level latent tensor modes may be more stable, transferable, and chemically interpretable than standard pooled embeddings. This paragraph should set up the paper’s core evidence packages: accuracy, stability, transfer, and intervention. ¹¹⁰

Paragraph 7. End with restrained claims. Avoid promising a new language of chemistry up front. Promise instead a reusable, symmetry-constrained response representation and say that any stronger interpretive language will be justified empirically in later sections.

Annotated bibliography and final verdict

The bibliography below is grouped by topic. I limited it to records whose metadata could be checked to reasonable confidence from publisher pages, arXiv, OpenReview, PubMed, PMLR, or equivalent stable sources during this review. A star marks **indispensable** references for writing the paper.

Annotated bibliography

Conceptual chemistry, representation, and interpretive framing

1. ★ **Atz, K.; Grisoni, F.; Schneider, G.** Geometric deep learning on molecular representations. **Nature Machine Intelligence** **2021**, 3, 1023–1032. DOI: [10.1038/s42256-021-00418-8](https://doi.org/10.1038/s42256-021-00418-8). [Peer-reviewed review.] A foundational review on molecular geometric deep learning, useful for positioning MTO-Net inside the existing field rather than as an entirely new one. **Supports** Narrative 3; **limits** claim 11 if overbranded. ¹¹¹
2. **Harnik, Y.; Specktor, E.; Cohen, Y.; Milo, A.; Pakman, A.** A Survey of Molecular Representation Learning: From Single Modalities to Foundation Models. **ChemRxiv** **2025**. DOI: [10.26434/chemrxiv-2025-vxsvs](https://doi.org/10.26434/chemrxiv-2025-vxsvs). [Preprint review.] Useful overview of where foundation-model and multimodal trends stand by 2025–2026. **Supports** taxonomy; **limits** claims of broad representation novelty. ¹¹²
3. **Reiser, P.; et al.** Graph neural networks for materials science and chemistry. **Communications Materials** **2022**. [Peer-reviewed review.] Broad review of graph neural networks in chemistry/materials; useful for grounding standard atom–bond/atomistic representation claims. **Supports** background; **limits** overclaiming about novelty of symmetry-aware chemistry ML. ¹¹³
4. **Galbraith, J. M.; et al.** Valence Bond and Molecular Orbital: Two Powerful Theories. **Journal of Chemical Education** **2021**. DOI: [10.1021/acs.jchemed.1c00919](https://doi.org/10.1021/acs.jchemed.1c00919). [Peer-reviewed.] Valuable for careful historical wording around VB/MO analogies. **Supports** a qualified analogy; **limits** triumphalist MO rhetoric. ¹¹⁴
5. **Shaik, S.** Valence Bond Theory—Its Birth, Struggles with Molecular Orbital Theory, and Future Perspectives. **2021**. [Peer-reviewed historical essay.] Useful for nuanced history and for avoiding a cartoon VB-vs-MO contrast. **Supports** historical background; **limits** simplistic phrasing. ¹¹⁵
6. ★ **Merino, G.; et al.** Aromaticity: Quo Vadis. **Chemical Science** **2023**. [Peer-reviewed perspective.] Excellent source on how core chemical concepts function despite definitional plurality. **Supports** Narrative 1 as reduced conceptual language; **limits** overly rigid concept claims. ¹¹⁶
7. **Solà, M.** Why Aromaticity Is a Suspicious Concept? Why? **Frontiers in Chemistry** **2017**. [Peer-reviewed review/perspective.] Shows that even central chemical concepts are theory-laden and multi-criterion. **Supports** cautious philosophical framing. ¹¹⁷
8. **Ertl, P.** The Most Common Functional Groups in Bioactive Molecules and How Their Popularity Has Evolved over Time. **Journal of Medicinal Chemistry** **2020**. DOI: [10.1021/acs.jmedchem.0c00754](https://doi.org/10.1021/acs.jmedchem.0c00754). [Peer-reviewed.] Concrete support for the centrality of functional-group reasoning in chemistry. **Supports** Narrative 1 and fragment-enrichment analysis design. ¹¹⁸

9. Li, Z.; et al. Review of group contribution methods for prediction ... **Molecular Physics** 2025. DOI: 10.1080/00268976.2025.2563020. [Peer-reviewed review.] Helpful for “computable intuition” precedents in reduced structural models. **Supports** Narrative 1. 119
10. Feng, F.; et al. Computational Chemical Synthesis Analysis and Pathway Design: Recent Advances and Future Directions. **WIREs Computational Molecular Science** / review context 2018. [Peer-reviewed review.] A good example of formalizing chemical expertise computationally. **Supports** the methodological, not philosophical, sense of computable intuition. 120

Equivariant and geometric molecular learning

1. ★ Thomas, N.; Smidt, T.; Kearnes, S.; Yang, L.; Li, L.; Kohlhoff, K.; Riley, P. Tensor Field Networks: Rotation- and Translation-Equivariant Neural Networks for 3D Point Clouds. **arXiv** 2018, arXiv:1802.08219. [Preprint.] Core irreps/tensor-field foundation. **Supports** Narrative 3. 48
2. Anderson, B.; Hy, T. S.; Kondor, R. Cormorant: Covariant Molecular Neural Networks. **NeurIPS** 2019. [Peer-reviewed conference.] Important early covariant molecular model. **Supports** background; **limits** novelty claims on equivariant molecular tensors. 49
3. Fuchs, F. B.; Worrall, D. E.; Fischer, V.; Welling, M. SE(3)-Transformers: 3D Roto-Translation Equivariant Attention Networks. **NeurIPS** 2020. [Peer-reviewed conference.] Key reference for equivariant attention and global 3D message passing. **Supports** background; **threatens** naive claims that global equivariant assembly is new. 50
4. Gasteiger, J.; Groß, J.; Günnemann, S. Directional Message Passing for Molecular Graphs. **ICLR** 2020. [Peer-reviewed conference.] Shows how modern graph models already move beyond purely local bondwise language. **Limits** overdrawn VB-like comparisons. 51
5. Gasteiger, J.; Becker, F.; Günnemann, S. GemNet: Universal Directional Graph Neural Networks for Molecules. **NeurIPS** 2021. [Peer-reviewed conference.] Relevant as a strong geometry-aware non-MTO baseline. **Limits** any claim that prior models lack diffuse/global chemistry. 52
6. ★ Schütt, K. T.; Unke, O. T.; Gastegger, M. Equivariant Message Passing for the Prediction of Tensorial Properties and Molecular Spectra. **ICML** 2021. [Peer-reviewed conference.] One of the most important background papers for MTO-Net. **Supports** equivariant tensor channels and response targets; **threatens** if MTO-Net does not go beyond atomwise equivariant representation plus readout. 53
7. ★ Batzner, S.; Musaelian, A.; Sun, L.; et al. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. **Nature Communications** 2022, 13, 2453. DOI: 10.1038/s41467-022-29939-5. [Peer-reviewed.] Major E(3)-equivariant benchmark. **Supports** symmetry background. 54
8. ★ Batatia, I.; Kovács, D. P.; Simm, G. N. C.; Ortner, C.; Csányi, G. MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields. **NeurIPS** 2022. [Peer-reviewed conference.] Strong higher-order equivariant backbone baseline. **Supports** local tensor-feature generation; **limits** novelty of backbone components. 55

9. **Musaelian, A.; Batzner, S.; Johansson, A.; et al.** Learning local equivariant representations for large-scale atomistic dynamics. **Nature Communications** **2023**. [Peer-reviewed.] Important for the distinction between rich local equivariant features and explicit global modes. **Supports** background. ¹²¹
10. **Liao, Y.-L.; Smidt, T.** Equiformer: Equivariant Graph Attention Transformer for 3D Atomistic Graphs. **ICLR 2023**. [Peer-reviewed conference.] Relevant to any comparison with global equivariant attention or multiple attention heads. **Threatens** weak distinctions from attention-style global structure. ⁵⁶
11. **Wang, Y.; Li, S.; He, X.; et al.** Enhancing geometric representations for molecules with equivariant vector-scalar interactive message passing. **Nature Communications** **2024**. [Peer-reviewed.] ViSNet is a strong response/property baseline and useful for irreps-aware probing comparisons. **Supports** baseline design. ⁵⁷
12. **Thölke, P.; De Fabritiis, G.** TorchMD-NET: Equivariant Transformers for Neural Network based Molecular Potentials. **arXiv 2022**, arXiv:2202.02541. [Preprint/software paper.] Notable because it already includes attention-weight analysis of learned geometric representations. **Limits** novelty if MTO-Net only offers qualitative attention-like interpretability. ¹²²

Tensorial properties, response, and spectroscopy

1. ★ **Grisafi, A.; Wilkins, D. M.; Csányi, G.; Ceriotti, M.** Symmetry-Adapted Machine Learning for Tensorial Properties of Atomistic Systems. **Physical Review Letters** **2018**, 120, 036002. DOI: **10.1103/PhysRevLett.120.036002**. [Peer-reviewed.] Foundational for the idea that tensor outputs require symmetry-respecting representations. **Supports** Narrative 3. ¹²³
2. ★ **Zou, Z.; Zhang, Y.; Liang, L.; Wei, M.; Leng, J.; Jiang, J.; Luo, Y.; Hu, W.** A deep learning model for predicting selected organic molecular spectra. **Nature Computational Science** **2023**, 3, 957–964. [Peer-reviewed.] One of the closest task-domain precedents. **Supports** spectroscopy framing; **threatens** if MTO-Net does not move beyond performance into representation science. ¹²⁴
3. ★ **Xu, Y.; Bian, D.; et al.** Pretrained E(3)-equivariant message-passing neural networks with multi-level representations for organic molecule spectra prediction. **npj Computational Materials** **2025**. [Peer-reviewed.] Probably the closest spectroscopy-side novelty threat because it already claims multi-level molecular representations under equivariant pretraining. **Threatens** MTO-Net unless explicit mode-bank distinctions are shown. ⁵⁹
4. ★ **Zhang, Y.; Jiang, B.** Universal machine learning for the response of atomistic systems to external fields. **Nature Communications** **2023**, 14, 6424. DOI: **10.1038/s41467-023-42148-y**. [Peer-reviewed.] Important precedent for response-oriented atomistic learning across dipole and polarizability-related observables. **Supports** Narrative 4 motivation; **limits** broad “response representation” novelty. ⁶⁰
5. ★ **Yan, W.; Lai, X.; Chen, Y.; Zhang, W.; Wu, J.; Xu, X.** General Framework for Geometric Deep Learning on Tensorial Properties of Molecules and Crystals. **Journal of the American Chemical Society** **2025**, 147, 47044–47056. DOI: **10.1021/jacs.5c12428**. [Peer-reviewed.] Very important for up-to-date tensorial-property methodology. **Supports** use of irreps and symmetry constraints; **threatens** broad framing around tensorial chemistry. ¹²⁵

6. **Gastegger, M.; Behler, J.; Marquetand, P.** Machine learning of solvent effects on molecular spectra and reactions. **Chemical Science** **2021**. [Peer-reviewed.] Useful for response/spectroscopy context and for arguing that hidden representations can capture perturbative response without being physical orbitals. **Supports** task relevance. ¹²⁶
7. **Hu, T.; Zou, Z.; Li, B.; Zhu, T.; Gu, S.; Jiang, J.; Luo, Y.; Hu, W.** Deep Learning for Bidirectional Translation between Molecular Structures and Vibrational Spectra. **Journal of the American Chemical Society** **2025**. DOI: [10.1021/jacs.5c05010](https://doi.org/10.1021/jacs.5c05010). [Peer-reviewed.] Useful recent spectroscopy prior art. **Supports** ambitious target space; **threatens** weak spectroscopy novelty claims. ¹²⁷
8. **Lee, G.; et al.** Machine-Learning Approach to Identify Organic Functional Groups of Compounds through Multiple Spectroscopic Inputs. **2025**. [Peer-reviewed.] Useful for connecting spectroscopy to functional-group-level interpretability. **Supports** fragment enrichment ideas. ¹²⁸

Electronic structure, orbitals, Hamiltonians, and orbital-informed ML

1. ★ **Schütt, K. T.; Gastegger, M.; Tkatchenko, A.; Müller, K.-R.; Maurer, R. J.** Unifying machine learning and quantum chemistry with a deep neural network for molecular wavefunctions. **Nature Communications** **2019**, 10, 5024. [Peer-reviewed.] Core closest-prior-art reference because it explicitly predicts wavefunction-related objects from atomistic inputs. **Threatens** any language suggesting MTOs are electronic structure. ⁶⁵
2. ★ **Unke, O. T.; Bogojeski, M.; Gastegger, M.; Geiger, M.; Smidt, T.; Müller, K.-R.** SE(3)-equivariant prediction of molecular wavefunctions and electronic densities. **NeurIPS** **2021**. [Peer-reviewed conference.] Strongest equivariant wavefunction-learning precedent in the molecular domain. **Threatens** overclaiming physical meaning. ⁶⁶
3. **Yu, H.; et al.** Efficient and Equivariant Graph Networks for Predicting Quantum Hamiltonian. **ICML** **2023**. [Peer-reviewed conference.] Important Hamiltonian-learning baseline. **Supports** distinction between latent response modes and operator prediction. ⁶⁷
4. **QH9 authors.** QH9: A Quantum Hamiltonian Prediction Benchmark for Molecules. **OpenReview/arXiv** **2023–2024**. [Benchmark paper.] Useful for benchmarking the Hamiltonian-learning category. **Supports** taxonomy and claim separation. ¹²⁹
5. **Zhong, Y.; et al.** Transferable equivariant graph neural networks for the Hamiltonians of molecules and solids. **npj Computational Materials** **2023**. DOI: [10.1038/s41524-023-01130-4](https://doi.org/10.1038/s41524-023-01130-4). [Peer-reviewed.] Stronger and more transferable Hamiltonian-learning prior art. **Threatens** any collapse of latent modes into physical Hamiltonian claims. ⁶⁸
6. **Wang, Z.; et al.** Infusing Self-Consistency into Density Functional Theory Hamiltonian Prediction via Deep Equilibrium Models. **arXiv / NeurIPS** **2024 context**. [Recent preprint.] Useful because it explicitly strengthens Hamiltonian learning using self-consistency. **Supports** contrast with non-electronic-structure learning. ¹³⁰
7. **Li, H.; Wang, Z.; Zou, N.; et al.** Deep-learning density functional theory Hamiltonian for efficient ab initio electronic-structure calculation. **Nature Computational Science** **2022**. [Peer-reviewed.] Materials-side but conceptually relevant for Hamiltonian prediction. **Supports** “what MTO-Net is not.” ¹³¹

8. **Shmilovich, K.; Willmott, D.; et al.** Orbital Mixer: Using Atomic Orbital Features for Basis-Dependent Prediction of Molecular Wavefunctions. **Journal of Chemical Theory and Computation** 2022. DOI: [10.1021/acs.jctc.2c00555](https://doi.org/10.1021/acs.jctc.2c00555). [Peer-reviewed.] Strong orbital-language prior art; useful for terminology boundaries. **Threatens** careless orbital branding. 69

9. **Qiao, Z.; Welborn, M.; Anandkumar, A.; Manby, F. R.; Miller, T. F. III.** OrbNet: Deep Learning for Quantum Chemistry Using Symmetry-Adapted Atomic-Orbital Features. **The Journal of Chemical Physics** 2020, 153, 124111. DOI: [10.1063/5.0021955](https://doi.org/10.1063/5.0021955). [Peer-reviewed.] Major orbital-informed ML precedent. **Threatens** novelty claims if MTO-Net allows extra quantum inputs. 132

10. **Christensen, A. S.; Sirumalla, S. K.; Qiao, Z.; O' Connor, M. B.; Smith, D. G. A.; et al.** OrbNet Denali: A machine learning potential for biological and organic chemistry with semi-empirical cost and DFT accuracy. **The Journal of Chemical Physics** 2021, 155, 204103. [Peer-reviewed.] Important for transfer and practical quantum-informed ML. **Supports** comparison tier. 133

11. ★ **Qiao, Z.; Christensen, A. S.; Welborn, M.; Manby, F. R.; Anandkumar, A.; Miller, T. F. III.** Informing geometric deep learning with electronic interactions to accelerate quantum chemistry. **PNAS** 2022. [Peer-reviewed.] Particularly relevant because it combines geometric deep learning with orbital/electronic interaction information. **Strong component overlap.** 134

12. ★ **CEONET authors.** Cartesian Equivariant Orbital Network / Cartesian equivariant representations for learning and predicting molecular orbitals and properties. **PNAS** 2025. [Peer-reviewed.] The most dangerous conceptual neighbor because it makes orbital-centered equivariant molecular representations explicit. **High threat** if MTO-Net uses sloppy orbital rhetoric. 135

13. ★ **Pfau, D.; Spencer, J. S.; Matthews, A. G. de G.; Foulkes, W. M. C.** Ab initio solution of the many-electron Schrödinger equation with deep neural networks. **Physical Review Research** 2020, 2, 033429. [Peer-reviewed.] Canonical FermiNet paper. **Supports** distinction from neural wavefunction methods. 71

14. ★ **Hermann, J.; Schätzle, Z.; Noé, F.** Deep-neural-network solution of the electronic Schrödinger equation. **Nature Chemistry** 2020, 12, 891–897. DOI: [10.1038/s41557-020-0544-y](https://doi.org/10.1038/s41557-020-0544-y). [Peer-reviewed.] Canonical PauliNet paper. **Supports** the same distinction. 72

15. **von Glehn, I.; Spencer, J. S.; Pfau, D.** A Self-Attention Ansatz for Ab-initio Quantum Chemistry. **ICLR 2023** / arXiv:2211.13672. [Peer-reviewed conference / preprint record.] Important for modern neural-wavefunction context. **Supports** what “neural orbitals” means in a physically distinct literature. 136

16. **Scherbela, M.; Gerard, L.; Grohs, P.** Towards a transferable fermionic neural wavefunction for molecules. **Nature Communications** 2024, 15, 120. DOI: [10.1038/s41467-023-44216-9](https://doi.org/10.1038/s41467-023-44216-9). [Peer-reviewed.] Important because it explicitly pursues transfer/reuse, but in true wavefunction space. **Supports** distinction and inspiration for transfer protocols. 137

Global, hierarchical, foundation, and chirality-related representations

1. ★ **Ying, R.; You, J.; Morris, C.; Ren, X.; Hamilton, W. L.; Leskovec, J.** Hierarchical Graph Representation Learning with Differentiable Pooling. **NeurIPS 2018**. [Peer-reviewed conference.]

The classic “maybe this is only pooling” threat. **Supports** baseline design; **threatens** shallow MTO claims. ⁶¹

2. ★ Baek, J.; Kang, M.; Hwang, S. J. Accurate Learning of Graph Representations with Graph Multiset Pooling. **ICLR 2021**. [Peer-reviewed conference.] Strong global-pooling baseline with multiple latent aggregation channels. **Direct threat** to claims that multiple global modes are automatically novel. ⁶²
3. ★ Han, J.; et al. Equivariant Graph Hierarchy-Based Neural Networks. **NeurIPS 2022**. [Peer-reviewed conference.] Closest architecture-side novelty threat because it already provides equivariant pooling/coarsening. **Strong component overlap**. ⁶³
4. ★ Zhou, G.; Gao, Z.; Ding, Q.; Zheng, H.; Xu, H.; Wei, Z.; Zhang, L.; Ke, G. Uni-Mol: A Universal 3D Molecular Representation Learning Framework. **ICLR 2023**. [Peer-reviewed conference.] Strong reusable 3D molecular representation baseline. **Supports** transfer comparisons; **limits** generic representation novelty. ⁷³
5. Liu, S.; Guo, H.; Tang, J. Pre-training Molecular Graph Representation with 3D Geometry. **ICLR 2022**. [Peer-reviewed conference.] Important precursor in 2D/3D cross-view pretraining. **Supports** pretraining context. ¹³⁸
6. Xia, J.; Zhao, H.; Liu, S.; et al. Rethinking Pre-training Graph Neural Networks for Molecules. **ICLR 2023**. [Peer-reviewed conference.] Mole-BERT is a strong data-driven molecular pretraining baseline. **Supports** discussion of reusable representations. ¹³⁹
7. ★ Jiao, R.; Kong, X.; Zhang, L.; Yu, Z.; Ren, F.; Tan, W.; Huang, W.; Liu, Y. An equivariant pretrained transformer for unified 3D molecular representation learning. **Nature Communications 2026**, 17, 2606. DOI: [10.1038/s41467-026-69185-7](https://doi.org/10.1038/s41467-026-69185-7). [Peer-reviewed.] Strongest recent foundation-model comparison because it combines pretraining and equivariant global molecular representations. **Threatens** weak reuse claims. ⁷⁴
8. Adams, K.; Pattanaik, L.; Coley, C. W. Learning 3D Representations of Molecular Chirality with Invariance to Bond Rotations. **ICLR 2022** / arXiv:2110.04383. [Peer-reviewed conference / preprint.] Important reference on chirality-sensitive representation learning. **Supports** symmetry-section caveats. ¹⁴⁰
9. Dumitrescu, A.; et al. E(3)-equivariant models cannot learn chirality. **arXiv / OpenReview 2024–2025**. [Recent preprint.] Important limit on naive symmetry claims. **Limits** any blanket “O(3) is always scientifically correct” wording. ¹⁴¹
10. Morehead, A.; Cheng, J. Geometry-Complete Perceptron Networks for 3D Molecular Graphs. **Bioinformatics 2024**, 40(2), btae087. DOI: [10.1093/bioinformatics/btae087](https://doi.org/10.1093/bioinformatics/btae087). [Peer-reviewed.] Useful chirality-aware geometric baseline. **Supports** nuanced chirality discussion. ¹⁴²

Chemical interpretability, concept learning, and representation stability

1. ★ Koh, P. W.; Nguyen, T.; Tang, Y. S.; et al. Concept Bottleneck Models. **ICML / PMLR 2020**. [Peer-reviewed conference.] Essential for any claim about computable high-level concepts and interventions. **Supports** Narrative 1 and 4 methods. ¹⁴³

2. **Abid, A.; et al.** Do Concept Bottleneck Models Respect Localities? **arXiv 2024**. [Preprint.] Important cautionary paper showing that concept methods can fail faithfulness tests. **Limits** concept-style interpretation claims. 144

3. ★ **Wu, Z.; et al.** Chemistry-intuitive explanation of graph neural networks for molecular property prediction with substructure masking. **Nature Communications 2023**. [Peer-reviewed.] One of the strongest chemistry-specific explanation precedents. **Supports** fragment enrichment and substructure analysis; **limits** mechanistic overclaiming. 145

4. **Proietti, M.; Ragno, A.; La Rosa, B.; et al.** Explainable AI in drug discovery: self-interpretable graph neural network for molecular property prediction using concept whitening. **Machine Learning 2024**. [Peer-reviewed.] Strong chemistry-specific concept-based interpretation baseline. **Supports** interpretable latent concepts. 146

5. **Aouichaoui, A. R. N.; Fan, F.; Mansouri, S. S.; Abildskov, J.; Sin, G.** Combining Group-Contribution Concept and Graph Neural Networks Toward Interpretable Molecular Property Models. **Journal of Chemical Information and Modeling 2023**, 63, 725–744. DOI: **10.1021/acs.jcim.2c01091**. [Peer-reviewed.] Directly relevant to fragment-level interpretable chemistry ML. **Supports** Narrative 1 and 4. 147

6. **Wellawatte, G. P.; Seshadri, A.; White, A. D.** A Perspective on Explanations of Molecular Prediction Models. **2023**. [Peer-reviewed perspective.] Clear warning against conflating explanation with mechanism. **Indispensable for claim hygiene**. 148

7. ★ **Sánchez-Lengeling, B.; Wei, J.; Lee, B.; et al.** Evaluating Attribution for Graph Neural Networks. **NeurIPS 2020**. [Peer-reviewed conference.] Gold-standard cautionary reference for graph attribution evaluation. **Limits** weak interpretability claims. 149

8. **Rao, J.; et al.** Quantitative evaluation of explainable graph neural networks for molecular property prediction. **Patterns 2022**. [Peer-reviewed.] Strong chemistry-specific explanation benchmark paper. **Limits** casual explanation claims; **supports** evaluation design. 150

9. **Raghu, M.; Gilmer, J.; Yosinski, J.; Sohl-Dickstein, J.** SVCCA: Singular Vector Canonical Correlation Analysis for Deep Learning Dynamics and Interpretability. **NeurIPS 2017**. [Peer-reviewed conference.] Useful for comparing hidden representations across seeds/layers. **Supports** claim-7 evaluation methods. 151

10. **Morcos, A. S.; Raghu, M.; Bengio, S.** Insights on representational similarity in neural networks with canonical correlation. **NeurIPS 2018**. [Peer-reviewed conference.] Projection-weighted CCA reference. **Supports** alignment analysis. 152

11. ★ **Kornblith, S.; Norouzi, M.; Lee, H.; Hinton, G.** Similarity of Neural Network Representations Revisited. **ICML 2019**. [Peer-reviewed conference.] The core CKA reference; essential for seed-stability and representation-comparison claims. 153

12. **Ding, F.; Denain, J.-S.; Steinhardt, J.** Grounding Representation Similarity Through Statistical Testing. **NeurIPS 2021**. [Peer-reviewed.] Important because it compares CKA, CCA, and orthogonal Procrustes and argues about specificity/sensitivity. **Supports** rigorous alignment choice. 154

13. **Knyazev, A. V.; Argentati, M. E.** Principal Angles between Subspaces in an A-Based Scalar Product. **SIAM Journal on Scientific Computing** 2002. [Peer-reviewed.] Classical principal-angle reference for comparing MTO subspaces. **Supports** claim-7 methodology. 155
14. **Klabunde, M.; Schumacher, L.; Strohmaier, M.; Lemmerich, F.** Similarity of Neural Network Models: A Survey of Functional and Representational Metrics. **ACM Computing Surveys** 2025. [Peer-reviewed survey.] Useful for not overselling any one alignment metric. 156
15. **Kivva, B.; Rajendran, G.; Ravikumar, P.; Aragam, B.** Identifiability of deep generative models without auxiliary information. **NeurIPS 2022**. [Peer-reviewed conference.] Helpful for clarifying that seed-stable subspaces are not automatically identifiable latent variables. **Limits** overclaiming on mode identity. 157
16. **Steier, J.; et al.** Information Routing in Atomistic Foundation Models. **arXiv** 2026. [Recent preprint.] Important novelty threat because it already studies frozen probes and irreps-specific routing in atomistic models. **Limits** claims that routing alone is new. 158

Final verdict

What is already known. The field already knows how to learn powerful atom-centered equivariant tensor representations; how to predict tensorial and spectroscopic properties; how to use orbital or Hamiltonian information when available; how to pretrain broad molecular representations; and how to produce fragment- or concept-level explanations, albeit imperfectly. None of those ingredients is individually new. 159

What appears genuinely new. The clearest open lane is an architecture that **explicitly assembles multiple molecule-level equivariant tensor modes from atom-centered tensor fields**, treats them as first-class latent objects, and demonstrates that they are **stable across seeds, reusable across response tasks, and more chemically interpretable than standard pooled embeddings**—all **without** electronic-structure supervision. I found no convincingly overlapping prior molecular paper that already offers that complete package. 160

What remains unproven. Nearly all stronger scientific claims in the proposed abstract remain unproven: that the modes are stable in a nontrivial sense, that they transfer beyond the training tasks, that they correspond to reusable chemical response concepts rather than arbitrary hidden coordinates, and that the MO analogy adds more than rhetoric. Those are empirical burdens for MTO-Net itself, not conclusions the literature already grants. 110

The exact scientific gap MTO-Net could fill. The most precise gap is this: **between atomwise equivariant hidden states and fully supervised electronic-structure objects there is still no widely established, explicitly parameterized, molecule-level equivariant latent response basis that is trained only from downstream response/spectroscopy supervision and then shown to be stable and reusable**. If MTO-Net fills that gap convincingly, it has a real claim. If it only improves readout, it does not. 161

Do the four narratives form a coherent paper? Yes, but only if they are weighted correctly. Narrative 1 motivates. Narrative 2 supplies architectural intuition with careful caveats. Narrative 3 supplies non-negotiable physical constraints. Narrative 4 carries the actual scientific contribution. The paper should be built to make Narrative 4 the evidentiary center of gravity. 162

How the abstract should be revised. The abstract should replace broad phrases like “makes inductive chemical intuition computable” with narrower, testable claims; qualify the VB/MO analogy as architectural inspiration; define MTOs immediately as **latent response modes, not orbitals**; and avoid claiming stable transfer or chemical meaning unless the paper contains serious stability, frozen-probe, and intervention evidence. A safer abstract would foreground: **explicit equivariant mode assembly, tensorial/spectroscopic prediction, multi-seed subspace reproducibility, cross-task response transfer, and controlled fragment-level analyses.** ¹⁶³

Open questions and limitations

The main unresolved issue after this review is not whether MTO-Net is interesting; it is whether its explicit molecule-level modes will survive skeptical comparisons to **equivariant pooling, global tokens, and recent atomistic-probing work**. Also unresolved is whether the term “response mode” will read as helpful or overloaded to spectroscopy/theoretical-chemistry audiences. Finally, because the literature through June 2026 is moving quickly—especially around equivariant pretrained models and probing analyses—future novelty drafts should be rechecked immediately before submission. ¹⁶⁴

简明中文执行摘要

总体判断：

MTO-Net 有机会成为一篇很强的化学机器学习 / AI for Science 论文，但前提是把它定位为：**一种把原子中心等变张量特征显式组装成分子级张量模式的表示学习框架**，而不是“神经分子轨道”“可计算化学直觉总理论”或“量子化学替代品”。其真正新意若存在，主要在于**显式分子级模式库 + 稳定性 + 可迁移性 + 可解释性证据**，而不是等变性本身。 ¹⁶⁵

最接近的先行工作：

若必须给出一个最接近方法，我会选 **CEONET**，因为它最接近“等变 + 轨道语言 + 分子级表示”这一组合；但 CEONET 学的是**真实轨道/轨道相关对象**，而 MTO-Net 设想的是**无电子结构监督的潜在响应模式**。在任务层面，**DetaNet / EnviroDetaNet** 是最接近的谱学与张量响应先行工作；在结构层面，**EGHN / DiffPool / GMT / EPT** 是最大的“这其实只是 pooling / token 机制”的新颖性威胁。 ¹⁶⁶

最大风险：

审稿人最可能的反对意见是：**MTO-Net 只是一个带有轨道修辞的等变 pooling / latent-slot / global-token 结构**。如果论文没有证明 MTO 具备：

1. 明确的分子级模式库；
2. 严格的 $O(3)$ 变换规律；
3. 跨随机种子稳定的子空间；
4. 跨任务可迁移；
5. 通过干预和反事实测试体现化学意义；

那么“新表示类”这个说法很难站住。 ¹⁶⁷

建议定位：

最安全的中心表述是：

“MTO-Net learns explicit delocalized molecule-level latent tensor modes from atom-centered equivariant features for tensorial and spectroscopic molecular prediction.”

中文可译为：

“MTO-Net 从原子中心等变张量特征中学习显式的、离域的、分子级潜在张量模式，用于张量性质与光谱性质预测。”

请避免直接说“神经分子轨道”“新的化学语言”“物理上有意义的轨道替代”。相反，应强调：**LCAO-inspired architectural analogy, not electronic-structure equivalence.** ¹⁶⁸

-
- 1 3 7 25 70 75 160 165 166 <https://www.osti.gov/pages/biblio/3005506>
<https://www.osti.gov/pages/biblio/3005506>
- 2 33 64 134 <https://www.pnas.org/doi/10.1073/pnas.2205221119>
<https://www.pnas.org/doi/10.1073/pnas.2205221119>
- 4 143 <https://proceedings.mlr.press/v119/koh20a.html>
<https://proceedings.mlr.press/v119/koh20a.html>
- 5 44 61 <https://arxiv.org/abs/1806.08804>
<https://arxiv.org/abs/1806.08804>
- 6 19 58 82 92 <https://pubmed.ncbi.nlm.nih.gov/38177591/>
<https://pubmed.ncbi.nlm.nih.gov/38177591/>
- 8 9 32 62 94 95 <https://openreview.net/forum?id=JHcqXGaqiGn>
<https://openreview.net/forum?id=JHcqXGaqiGn>
- 10 74 <https://pubmed.ncbi.nlm.nih.gov/41667467/>
<https://pubmed.ncbi.nlm.nih.gov/41667467/>
- 11 12 23 24 80 108 114 163 168 <https://pubs.acs.org/doi/10.1021/acs.jchemed.1c00919>
<https://pubs.acs.org/doi/10.1021/acs.jchemed.1c00919>
- 13 <https://orbit.dtu.dk/en/publications/molecular-response-properties-in-equation-of-motion-coupled-clust>
<https://orbit.dtu.dk/en/publications/molecular-response-properties-in-equation-of-motion-coupled-clust>
- 14 15 110 153 167 <https://arxiv.org/abs/1905.00414>
<https://arxiv.org/abs/1905.00414>
- 16 63 93 164 https://papers.neurips.cc/paper_files/paper/2022/file/3bdeb28a531f7af94b56bcdf8ee88f17-Paper-Conference.pdf
https://papers.neurips.cc/paper_files/paper/2022/file/3bdeb28a531f7af94b56bcdf8ee88f17-Paper-Conference.pdf
- 17 20 22 30 36 106 111 162 https://www.researchgate.net/publication/357082576_Geometric_deep_learning_on_molecular_representations
https://www.researchgate.net/publication/357082576_Geometric_deep_learning_on_molecular_representations
- 18 59 <https://www.nature.com/articles/s41524-025-01698-z>
<https://www.nature.com/articles/s41524-025-01698-z>
- 21 76 119 <https://www.tandfonline.com/doi/full/10.1080/00268976.2025.2563020>
<https://www.tandfonline.com/doi/full/10.1080/00268976.2025.2563020>
- 26 40 48 <https://arxiv.org/abs/1802.08219>
<https://arxiv.org/abs/1802.08219>
- 27 41 65 81 84 96 104 161 <https://www.nature.com/articles/s41467-019-12875-2>
<https://www.nature.com/articles/s41467-019-12875-2>
- 28 101 141 <https://arxiv.org/html/2402.15864v2>
<https://arxiv.org/html/2402.15864v2>
- 29 77 144 <https://arxiv.org/html/2401.01259v2>
<https://arxiv.org/html/2401.01259v2>

31 51 79 <https://arxiv.org/abs/2003.03123>
<https://arxiv.org/abs/2003.03123>

34 45 98 107 123 <https://link.aps.org/doi/10.1103/PhysRevLett.120.036002>
<https://link.aps.org/doi/10.1103/PhysRevLett.120.036002>

35 39 103 158 <https://arxiv.org/html/2603.03155v1>
<https://arxiv.org/html/2603.03155v1>

37 87 102 151 <https://arxiv.org/abs/1706.05806>
<https://arxiv.org/abs/1706.05806>

38 46 90 145 <https://www.nature.com/articles/s41467-023-38192-3>
<https://www.nature.com/articles/s41467-023-38192-3>

42 <https://arxiv.org/abs/2007.08026>
<https://arxiv.org/abs/2007.08026>

43 71 <https://link.aps.org/doi/10.1103/PhysRevResearch.2.033429>
<https://link.aps.org/doi/10.1103/PhysRevResearch.2.033429>

47 73 89 <https://openreview.net/forum?id=6K2RM6wVqKu>
<https://openreview.net/forum?id=6K2RM6wVqKu>

49 <https://arxiv.org/abs/1906.04015>
<https://arxiv.org/abs/1906.04015>

50 <https://deepai.org/publication/se-3-transformers-3d-roto-translation-equivariant-attention-networks>
<https://deepai.org/publication/se-3-transformers-3d-roto-translation-equivariant-attention-networks>

52 <https://proceedings.neurips.cc/paper/2021/hash/35cf8659cfcb13224cbd47863a34fc58-Abstract.html>
<https://proceedings.neurips.cc/paper/2021/hash/35cf8659cfcb13224cbd47863a34fc58-Abstract.html>

53 85 100 109 159 <https://proceedings.mlr.press/v139/schutt21a/schutt21a.pdf>
<https://proceedings.mlr.press/v139/schutt21a/schutt21a.pdf>

54 <https://www.nature.com/articles/s41467-022-29939-5>
<https://www.nature.com/articles/s41467-022-29939-5>

55 https://proceedings.neurips.cc/paper_files/paper/2022/hash/4a36c3c51af11ed9f34615b81edb5bbc-Abstract-Conference.html
https://proceedings.neurips.cc/paper_files/paper/2022/hash/4a36c3c51af11ed9f34615b81edb5bbc-Abstract-Conference.html

56 <https://arxiv.org/abs/2206.11990>
<https://arxiv.org/abs/2206.11990>

57 <https://www.nature.com/articles/s41467-023-43720-2>
<https://www.nature.com/articles/s41467-023-43720-2>

60 <https://www.nature.com/articles/s41467-023-42148-y>
<https://www.nature.com/articles/s41467-023-42148-y>

66 <https://proceedings.neurips.cc/paper/2021/file/78f1893678afbeaa90b1fa01b9cfb860-Paper.pdf>
<https://proceedings.neurips.cc/paper/2021/file/78f1893678afbeaa90b1fa01b9cfb860-Paper.pdf>

67 <https://proceedings.mlr.press/v202/yu23i/yu23i.pdf>
<https://proceedings.mlr.press/v202/yu23i/yu23i.pdf>

68 <https://www.nature.com/articles/s41524-023-01130-4>
<https://www.nature.com/articles/s41524-023-01130-4>

69 <https://pubs.acs.org/doi/10.1021/acs.jctc.2c00555>
<https://pubs.acs.org/doi/10.1021/acs.jctc.2c00555>

72 <https://www.nature.com/articles/s41557-020-0544-y>
<https://www.nature.com/articles/s41557-020-0544-y>

78 113 <https://www.nature.com/articles/s43246-022-00315-6>
<https://www.nature.com/articles/s43246-022-00315-6>

83 86 <https://arxiv.org/abs/2106.08903>
<https://arxiv.org/abs/2106.08903>

88 157 https://proceedings.neurips.cc/paper_files/paper/2022/file/649f080d8891ab4d4b262cb9cd52e69a-Paper-Conference.pdf
https://proceedings.neurips.cc/paper_files/paper/2022/file/649f080d8891ab4d4b262cb9cd52e69a-Paper-Conference.pdf

91 97 148 <https://pmc.ncbi.nlm.nih.gov/articles/PMC10134429/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10134429/>

99 <https://pubs.acs.org/doi/10.1021/acscatal.5b00007>
<https://pubs.acs.org/doi/10.1021/acscatal.5b00007>

105 118 <https://pubs.acs.org/doi/abs/10.1021/acs.jmedchem.0c00754>
<https://pubs.acs.org/doi/abs/10.1021/acs.jmedchem.0c00754>

112 https://www.researchgate.net/publication/391727971_A_Survey_of_Molecular_Representation_Learning_From_Single_Modalities_to_Foundation_Models
https://www.researchgate.net/publication/391727971_A_Survey_of_Molecular_Representation_Learning_From_Single_Modalities_to_Foundation_Models

115 <https://pmc.ncbi.nlm.nih.gov/articles/PMC8001733/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8001733/>

116 <https://pubs.rsc.org/en/content/articlehtml/2023/sc/d2sc04998h>
<https://pubs.rsc.org/en/content/articlehtml/2023/sc/d2sc04998h>

117 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5364188/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5364188/>

120 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5994992/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5994992/>

121 <https://www.nature.com/articles/s41467-023-36329-y>
<https://www.nature.com/articles/s41467-023-36329-y>

122 <https://arxiv.org/abs/2202.02541>
<https://arxiv.org/abs/2202.02541>

124 <https://www.chemsoc.org.cn/member/senior/89936.html>
<https://www.chemsoc.org.cn/member/senior/89936.html>

125 <https://xdft.fudan.edu.cn/24466/list.htm>
<https://xdft.fudan.edu.cn/24466/list.htm>

126 <https://pubs.rsc.org/en/content/articlehtml/2021/sc/d1sc02742e>
<https://pubs.rsc.org/en/content/articlehtml/2021/sc/d1sc02742e>

127 <https://pubs.acs.org/doi/10.1021/jacs.5c05010>
<https://pubs.acs.org/doi/10.1021/jacs.5c05010>

- 128 <https://pmc.ncbi.nlm.nih.gov/articles/PMC11966250/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11966250/>
- 129 <https://arxiv.org/html/2306.09549v4>
<https://arxiv.org/html/2306.09549v4>
- 130 <https://arxiv.org/abs/2406.03794>
<https://arxiv.org/abs/2406.03794>
- 131 <https://www.nature.com/articles/s43588-022-00265-6>
<https://www.nature.com/articles/s43588-022-00265-6>
- 132 <https://pubmed.ncbi.nlm.nih.gov/33003742/>
<https://pubmed.ncbi.nlm.nih.gov/33003742/>
- 133 <https://pubs.aip.org/aip/jcp/article/155/20/204103/280547/OrbNet-Denali-A-machine-learning-potential-for>
<https://pubs.aip.org/aip/jcp/article/155/20/204103/280547/OrbNet-Denali-A-machine-learning-potential-for>
- 135 <https://www.pnas.org/doi/10.1073/pnas.2510235122>
<https://www.pnas.org/doi/10.1073/pnas.2510235122>
- 136 <https://arxiv.org/abs/2211.13672>
<https://arxiv.org/abs/2211.13672>
- 137 <https://www.nature.com/articles/s41467-023-44216-9>
<https://www.nature.com/articles/s41467-023-44216-9>
- 138 <https://arxiv.org/abs/2110.07728>
<https://arxiv.org/abs/2110.07728>
- 139 <https://openreview.net/forum?id=jevY-DtiZTR>
<https://openreview.net/forum?id=jevY-DtiZTR>
- 140 <https://arxiv.org/abs/2110.04383>
<https://arxiv.org/abs/2110.04383>
- 142 <https://pubmed.ncbi.nlm.nih.gov/38373819/>
<https://pubmed.ncbi.nlm.nih.gov/38373819/>
- 146 <https://link.springer.com/article/10.1007/s10994-023-06369-y>
<https://link.springer.com/article/10.1007/s10994-023-06369-y>
- 147 <https://pubs.acs.org/doi/10.1021/acs.jcim.2c01091>
<https://pubs.acs.org/doi/10.1021/acs.jcim.2c01091>
- 149 <https://papers.nips.cc/paper/2020/file/417fbbf2e9d5a28a855a11894b2e795a-Paper.pdf>
<https://papers.nips.cc/paper/2020/file/417fbbf2e9d5a28a855a11894b2e795a-Paper.pdf>
- 150 <https://www.cell.com/patterns/fulltext/S2666-3899%2822%2900260-4>
<https://www.cell.com/patterns/fulltext/S2666-3899%2822%2900260-4>
- 152 <https://papers.nips.cc/paper/7815-insights-on-representational-similarity-in-neural-networks-with-canonical-correlation>
<https://papers.nips.cc/paper/7815-insights-on-representational-similarity-in-neural-networks-with-canonical-correlation>
- 154 https://openreview.net/forum?id=_kwj6V53ZqB
https://openreview.net/forum?id=_kwj6V53ZqB

¹⁵⁵ <https://epubs.siam.org/doi/10.1137/S1064827500377332>

<https://epubs.siam.org/doi/10.1137/S1064827500377332>

¹⁵⁶ <https://dl.acm.org/doi/full/10.1145/3728458>

<https://dl.acm.org/doi/full/10.1145/3728458>